EDA

2023-03-03

Exploratory_Data_Analysis - Melbourne Housing Prices

To create a PDF output, make sure that you have the latex installed. You need to run tinytex::install_tinytex()

Q1. Read the file and print summary

```
library(tidyverse)
```

```
## -- Attaching packages --------------------------------------- tidyverse 1.3.2 --
## v ggplot2 3.4.1     v purrr   1.0.1
## v tibble  3.1.8     v dplyr   1.1.0
## v tidyr   1.3.0     v stringr 1.5.0
## v readr   2.1.4     v forcats 1.0.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
```

```
data <- read_csv("melbourne_housing_prices.csv", show_col_types = FALSE)

summary(data)
```

```
##     Suburb            Address              Rooms           Type
##  Length:13580       Length:13580       Min.   : 1.000   Length:13580
##  Class :character   Class :character   1st Qu.: 2.000   Class :character
##  Mode  :character   Mode  :character   Median : 3.000   Mode  :character
##                                        Mean   : 2.938
##                                        3rd Qu.: 3.000
##                                        Max.   :10.000
##
##      Price            Method            SellerG             Date
##  Min.   :  85000   Length:13580       Length:13580       Length:13580
##  1st Qu.: 650000   Class :character   Class :character   Class :character
##  Median : 903000   Mode  :character   Mode  :character   Mode  :character
##  Mean   :1075684
##  3rd Qu.:1330000
##  Max.   :9000000
##
##     Distance        Postcode       Bedroom2         Bathroom
##  Min.   : 0.00   Min.   :3000   Min.   : 0.000   Min.   :0.000
##  1st Qu.: 6.10   1st Qu.:3044   1st Qu.: 2.000   1st Qu.:1.000
##  Median : 9.20   Median :3084   Median : 3.000   Median :1.000
##  Mean   :10.14   Mean   :3105   Mean   : 2.915   Mean   :1.534
##  3rd Qu.:13.00   3rd Qu.:3148   3rd Qu.: 3.000   3rd Qu.:2.000
##  Max.   :48.10   Max.   :3977   Max.   :20.000   Max.   :8.000
```

```
##
##       Car            Landsize         BuildingArea     YearBuilt
##  Min.   : 0.00   Min.   :     0.0   Min.   :    0   Min.   :1196
##  1st Qu.: 1.00   1st Qu.:   177.0   1st Qu.:   93   1st Qu.:1940
##  Median : 2.00   Median :   440.0   Median :  126   Median :1970
##  Mean   : 1.61   Mean   :   558.4   Mean   :  152   Mean   :1965
##  3rd Qu.: 2.00   3rd Qu.:   651.0   3rd Qu.:  174   3rd Qu.:1999
##  Max.   :10.00   Max.   :433014.0   Max.   :44515   Max.   :2018
##  NA's   :62                         NA's   :6450    NA's   :5375
##  CouncilArea          Lattitude        Longtitude     Regionname
##  Length:13580       Min.   :-38.18   Min.   :144.4   Length:13580
##  Class :character   1st Qu.:-37.86   1st Qu.:144.9   Class :character
##  Mode  :character   Median :-37.80   Median :145.0   Mode  :character
##                     Mean   :-37.81   Mean   :145.0
##                     3rd Qu.:-37.76   3rd Qu.:145.1
##                     Max.   :-37.41   Max.   :145.5
##
##  Propertycount
##  Min.   :  249
##  1st Qu.: 4380
##  Median : 6555
##  Mean   : 7454
##  3rd Qu.:10331
##  Max.   :21650
##
```

Q2. Which variables/vectors have missing data

```
print(colSums(is.na(data)))
```

```
##        Suburb       Address         Rooms          Type         Price
##             0             0             0             0             0
##        Method       SellerG          Date      Distance      Postcode
##             0             0             0             0             0
##      Bedroom2      Bathroom           Car      Landsize  BuildingArea
##             0             0            62             0          6450
##     YearBuilt   CouncilArea     Lattitude    Longtitude    Regionname
##          5375          1369             0             0             0
## Propertycount
##             0
```

```
print("--------------Percentage Of missing values--------------")
```

```
## [1] "--------------Percentage Of missing values--------------"
```

```
# percentage of missing value
print(colSums(is.na(data))*100/nrow(data))
```

```
##        Suburb       Address         Rooms          Type         Price
##     0.0000000     0.0000000     0.0000000     0.0000000     0.0000000
##        Method       SellerG          Date      Distance      Postcode
##     0.0000000     0.0000000     0.0000000     0.0000000     0.0000000
##      Bedroom2      Bathroom           Car      Landsize  BuildingArea
##     0.0000000     0.0000000     0.4565538     0.0000000    47.4963181
##     YearBuilt   CouncilArea     Lattitude    Longtitude    Regionname
##    39.5802651    10.0810015     0.0000000     0.0000000     0.0000000
```

```
## Propertycount
##     0.0000000
```

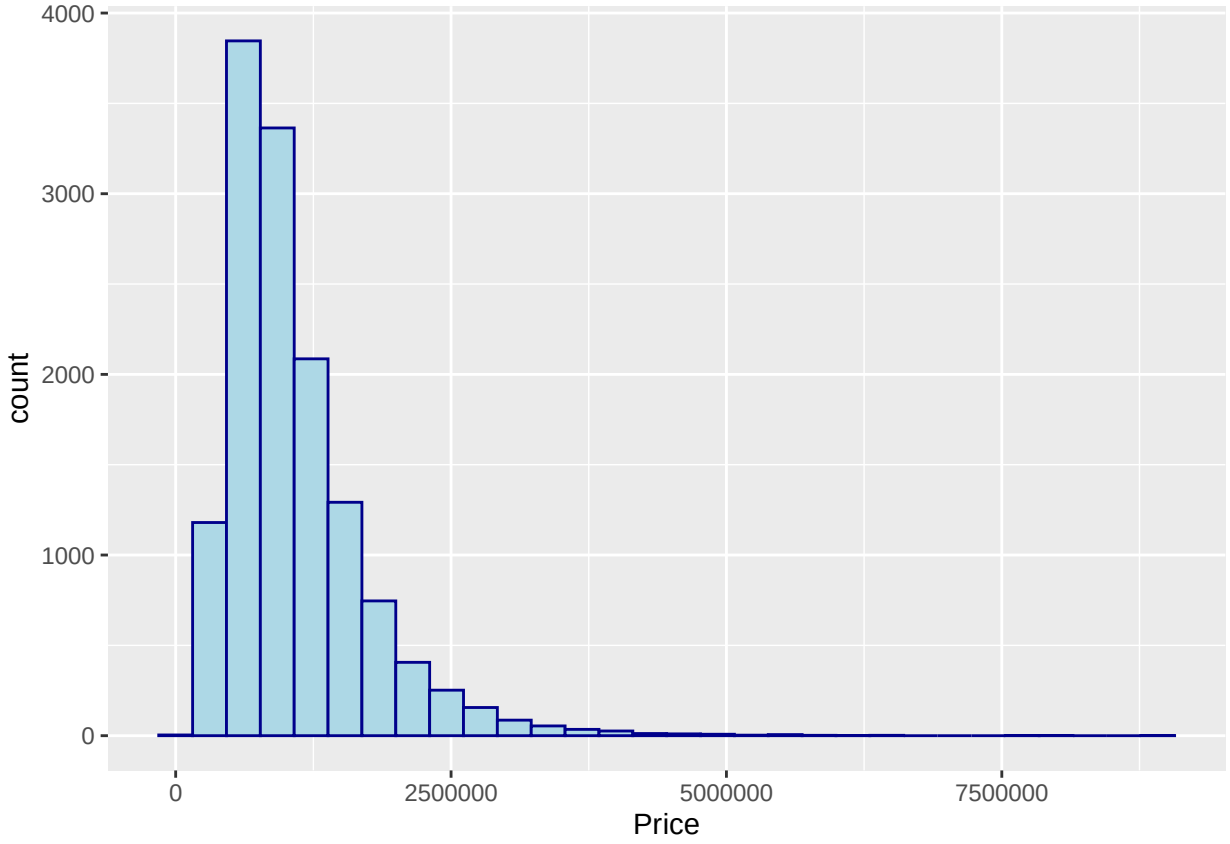
Here are the columns that have missing values are: car, CounciArea, BuildingArea and YearBuilt.

It would be more interesting to remove the BuildingArea and YearBuilt columns. Since each of these columns has respectively 47.49% and 39.58% of missing value. However, for the variables car and CouncilArea we can fill in the missing values by the mean.

How is the price distributed? Create a histogram using ggplot2

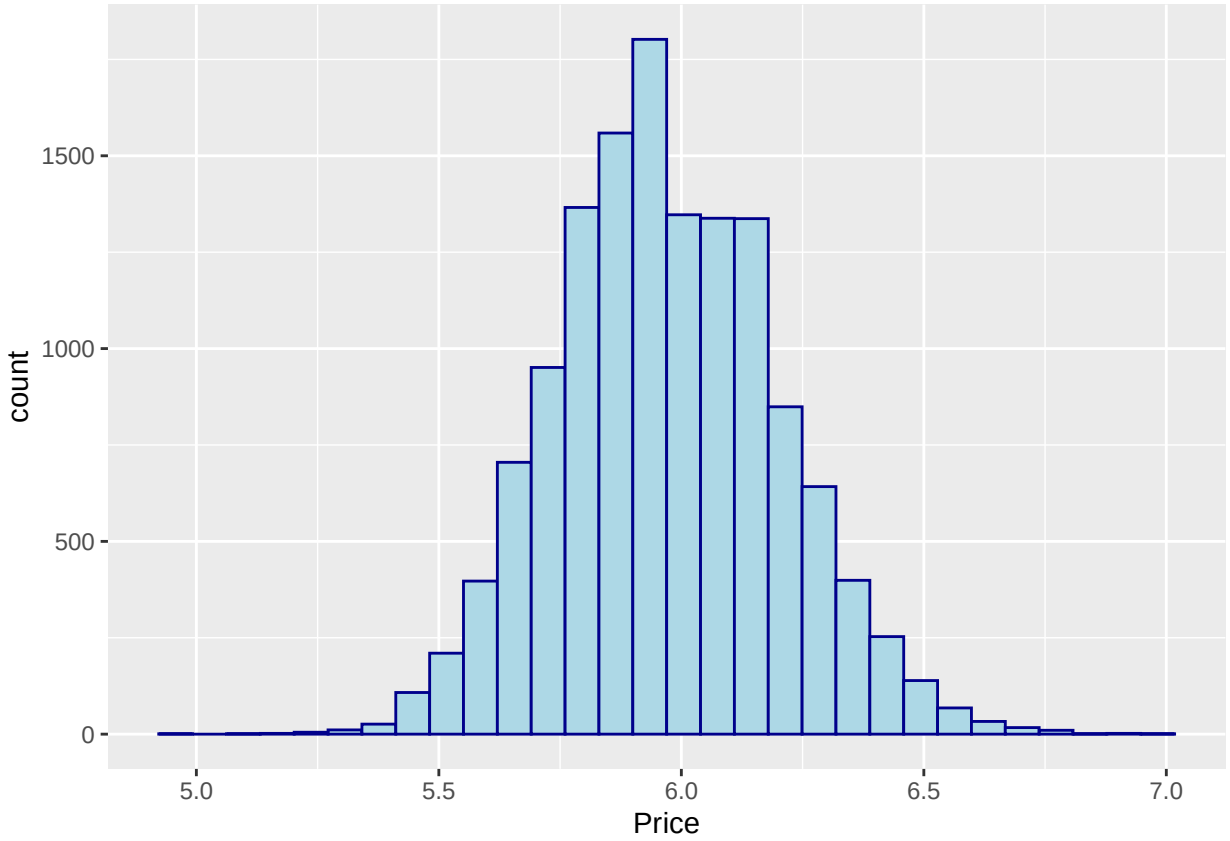
```
library(ggplot2)

ggplot(data, aes(x=Price))+
  geom_histogram(color="darkblue", fill="lightblue", bins = 30)
```




```
data["log10_Price"]=log10(data$Price)
library(ggplot2)

ggplot(data, aes(x=log10_Price))+
  geom_histogram(color="darkblue", fill="lightblue", bins = 30)+
  labs(x="Price")
```



Q4. Will the number of bedrooms impact the price of the house?

```
par(mfrow=c(3,1))
```

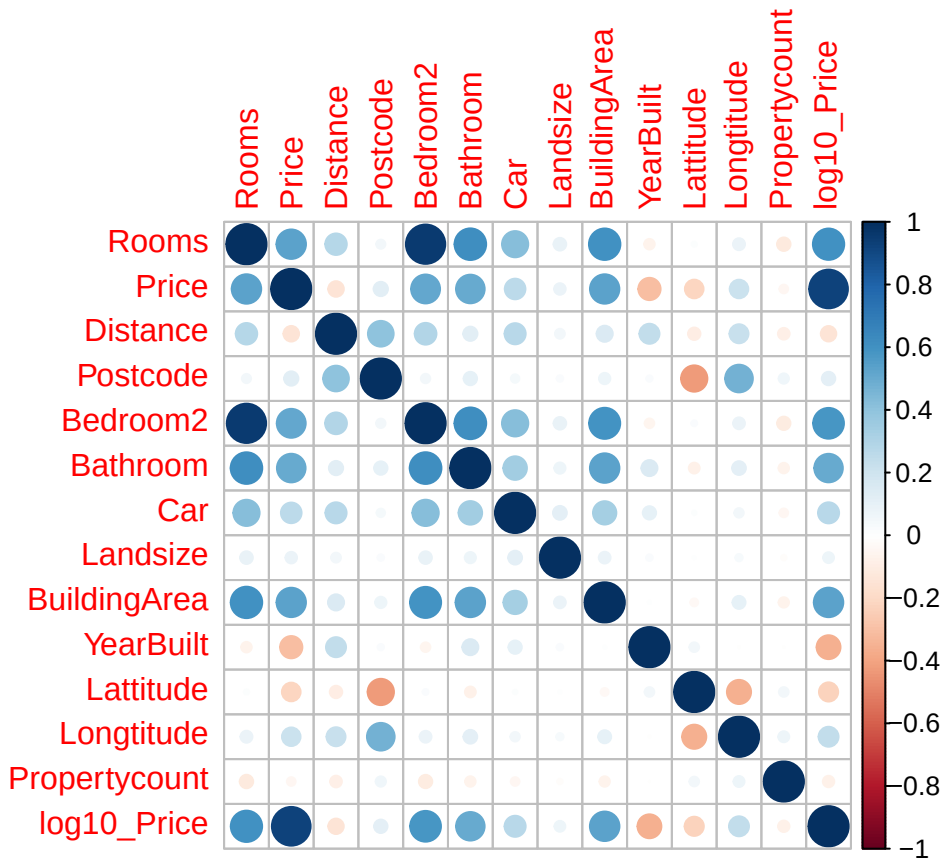
Create a correlation matrix for the numeric variables included below (Do not include any character variable for correlation analysis) and display the correlation matrix. Additional information is provided for your reference. INTERPRET THE CORRELATION ANALYSIS IN YOUR OWN WORDS AND IN A DETAILED MANNER TO RECEIVE FULL CREDIT FOR THIS QUESTION

```
library(corrplot)
```

```
## corrplot 0.92 loaded
```

```
data <- na.omit(data)
M = cor(select(data, -c("Suburb","Address","Type","Method", "SellerG", "Date", "CouncilArea", "Regionnar


corrplot(M, method = "circle")
```


Rooms
Price
Distance
Postcode
Bedroom2
Bathroom
Car
Landsize
BuildingArea
YearBuilt
Lattitude
Longtitude
Propertycount
log10_Price
1
0.8
0.6
0.4
0.2
0
−0.2
−0.4
−0.6
−0.8
−1